Tutorial 6 - More SQL

Please go through the chapter 7, if you haven't already, yes initially reading and understanding the code to learn about query concepts (e.g. there many types of subqueries discussed) is also fine. After the concepts' discussion towards the end of chapter the book is discussing Crafting SELECT Queries (7-11, pg. 317), which tells you how to approach query writing. Read and apply that while writing the queries. The screen shots provided for each query are helpful as they guide you what is the expected query set. There are hints and enough guidance in the query questions.

"Modify some of the query results (for formatting purposes) based on the Notes given at the end of Tutorial 5."

> Compare the results closely especially amounts and decimal places and see if you can format your query results. You can use inbuilt functions (e.g. to_char (), cast, etc. that are part of the chapter 7.8 to 7.13 – make sure to use Oracle specific formatting) or format command that is slightly old-fashioned and must be applied for each login session.

Once again, practising by doing is the mantra to learn, can't emphasise enough. If you get stuck and have tried googling based on the oracle error code, use the course forum to post your query (along with Tutorial no and question, better to provide both the **screen shots and soft copy of code** as well). Questions via email are fine, but better if posted on forum as then you can help each other and many of you might have similar issues. Sending code via email to me is also fine but consider the forum option first. Thanks for your understanding.

Task 1: General comments:

Many of the questions are building from the earlier question and require you to further modify the query code. Suggest follow the given order of the questions, but if get stuck at times it helps to leave and move forward and then revisit :)
Where screen shots of the query result are not provided, try to get the answer via looking at the data, and this should be possible and not too onerous as the database is small. This reinforces the understanding of the question itself as well. The document will also indicate some information against the question where it is missing or difficult to build.

Task 2: General comments:

Very large database takes a while to run the script file. There are some inserts that were commented as they were violating the create table constraints. Some create table constraints were also modified to improve the data quality. The queries are not too difficult especially after doing Tutorial 5 and Tutorial 6 Task1 queries. The screen shots provided for each query are helpful as they guide you what is the expected query set. For some questions, the screen shots display limited rows and not complete results (because a query might return 100s of records). The document lists the questions that don't have complete query results and indicate the number of records.

Task1 Q9-26

Run the code from the script file Ch07SaleCo.sql provided on Stream to create and populate the following five tables (CUSTOMER, INVOICE, LINE, PRODUCT, VENDOR).

9) 8 invoices in the Invoice table
10) 2 customers
11) Only use the required tables. Although you could write a correct query if you include Customer table but shouldn't because this is unnecessary. Invoice table has customer number that we need, and for this query result we do not require additional customer information (maintenance and other issues with additional code). Consider applying function to match the date format as in the screen shot.
12) See if you can change the column headings to match by using aliases. Results are for the 5 customers (10011,10012,10014,10015 and 10018) who made purchases.
13) Tables used: customer, invoice and line; Need to use Group By.
14) Just need to add another column (count(*)) to Q13
15) Need to include Avg now.
16) Requires single table only.
17) –
18) Need to eliminate duplicates for an aggregate function
19) Apply aggregate to aggregate [e.g. sum(count())]
20) May consider using a subquery (e.g. IN – 7.8b) //just to get some practice, if not already used in your earlier work.
21) Try to match columns names (e.g. Minimum Balance by using aliases). How will you take care of two words with a space?
22) Just need to add another column (from Q21)
23) Like Q21 (just need to negate – who didn't make purchases). May use an appropriate logical operator.
24) Building on from previous using aggregate functions.
25) Need to use single table only.
26) Apply aggregate to one of the columns from the above query.

Task2 (Some more suggested SQL queries for practice) Q27-44

Run the code from the script file Ch07LargeCo.sql to create and populate the following 10 tables (LGCUSTOMER, LGINVOICE, LGLINE, LGPRODUCT, LGBRAND, LGSUPPLIES, LGSALARY_HISTORY, LGEMPLOYEE, LGDEPARTMENT, LGVENDOR).

27) –
28) What if you wanted to display Product description up to 25 characters only out of 120 possible (as per table structure)? What function could you use?
29) 104 rows should be returned as per the given database.
30) 31 rows returned
31) –
32) No rows selected – don't worry your query might be correct;
25 displayed rows are returned if you change the date range to July 15, 2017, and July 31, 2017 (i.e. book typo – year not 2015)
The question also doesn't make following very clear:
The brand_name is 'FORESTERS BEST' and prod_category is 'Top Coat' – both case sensitive.
33) 168 rows selected.
Displayed ordering of the query is different//but records match
Job title in question is obviously referring to EMP TITLE
34) –
35) –
36) –
37) –
38) Pay attention to the rounded decimal requirements. You can play with changing the value (say 1 from 2 and see how results change).
39) How will you change date format to display month - mm before the date - dd?
40) 33 rows selected.
41) 43 rows selected. Need to apply criteria to filter records after grouping (based on invoice total).
42) –
43) 139 rows selected.
44) ……keep going if you wish 😊